

**A PROJECT ON
“RECIRCLE- MART WEB APPLICATION”**

SUBMITTED IN
PARTIAL FULFILLMENT OF THE REQUIREMENT
FOR THE COURSE OF DIPLOMA IN ADVANCED COMPUTING FROM CDAC



SUNBEAM INSTITUTE OF INFORMATION TECHNOLOGY
Hinjawadi

SUBMITTED BY:
Prashant Kumar
Rohit Kavathekar
Harsh Kumar
Om Dhavale

UNDER THE GUIDENCE OF:
Vrushabh Kumatale
Faculty Member
Sunbeam Institute of Information Technology, Pune

ACKNOWLEDGEMENT

A project usually falls short of its expectation unless aided and guided by the right persons at the right time. We avail this opportunity to express our deep sense of gratitude towards Mr. Nitin Kudale (Center Coordinator, SIIT, Pune) and Mr. Yogesh Kolhe (Course Coordinator, SIIT ,Pune) .

We are deeply indebted and grateful to them for their guidance, encouragement and deep concern for our project. Without their critical evaluation and suggestions at every stage of the project, this project could never have reached its present form.

Last but not the least we thank the entire faculty and the staff members of Sunbeam Institute of Information Technology, Pune for their support.

Prashant Kumar
Rohit Kavathekar
Harsh Kumar
Om Dhavale
0825 PG-DAC
SIIT Pune

1.INTRODUCTION TO PROJECT

The **Recircle Mart Web Application** is a web-based platform designed to promote **reuse, resale, and recycling of products** by connecting sellers and buyers on a single digital marketplace. The system aims to reduce waste, support sustainability, and encourage a circular economy by allowing users to list, browse, buy, and manage second-hand or recyclable products efficiently.

The application provides an easy-to-use interface where users can **search for products, compare prices, view product details, and purchase items securely**. Users can access basic features such as browsing and searching products without logging in. However, to **sell products, place orders, or manage profiles**, users must register and log into the system.

Recircle Mart allows sellers to list products with details such as **product name, category, condition, price, images, and description**. Buyers can explore products by category, location, and price range. The system ensures transparency by displaying seller details, product availability, and pricing information clearly.

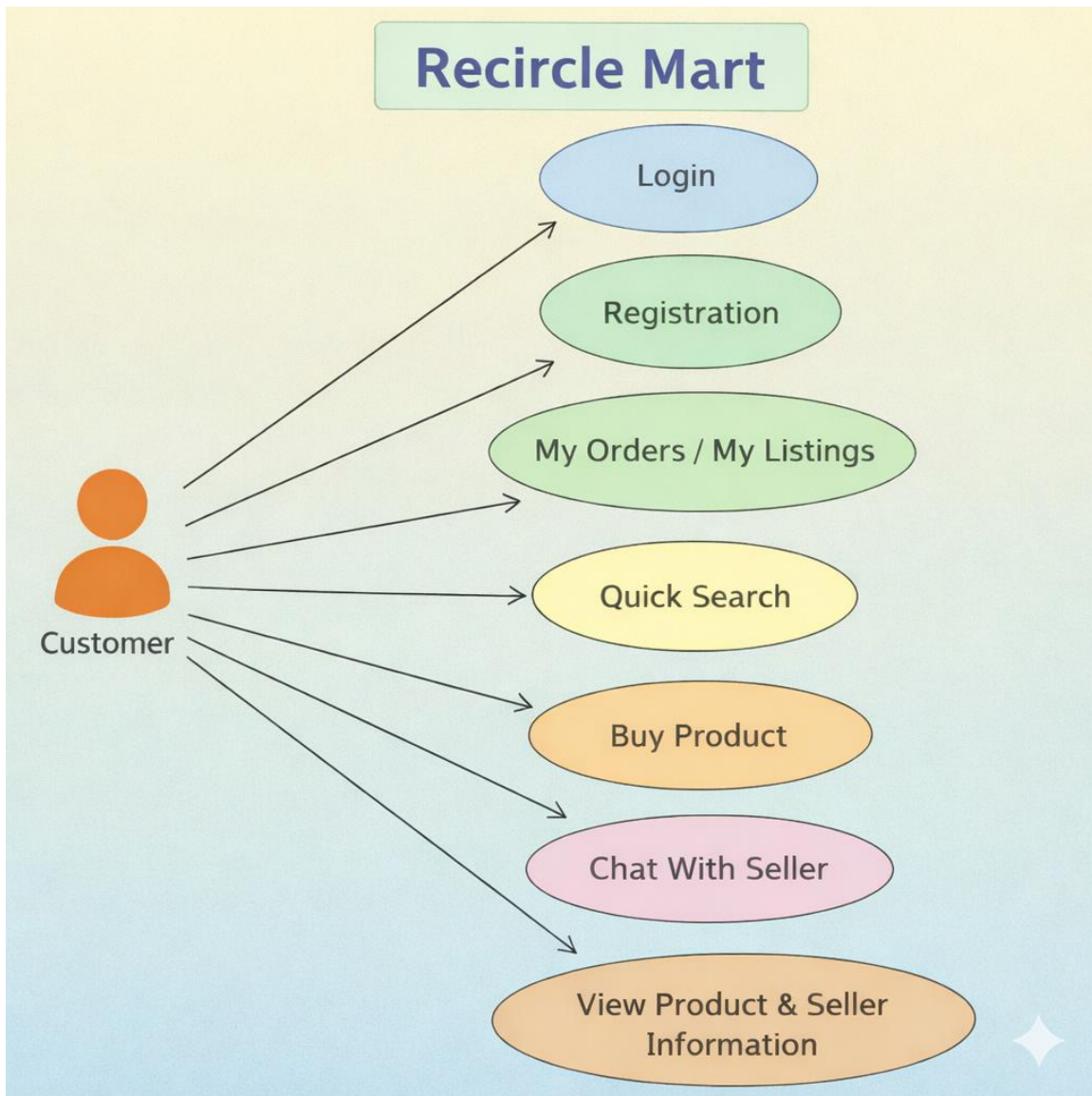
The platform also includes an **admin module** that allows administrators to manage users, product listings, categories, and reported items. Admins ensure smooth operation, data integrity, and policy compliance across the system.

By digitizing the resale and recycling process, Recircle Mart provides a structured, secure, and eco-friendly solution that benefits individuals, businesses, and the environment.

-
- **Key Objectives of Recircle Mart**
 - Promote reuse and recycling of products
 - Reduce environmental waste
 - Provide a secure online resale platform
 - Enable easy product listing and discovery
 - Support sustainable buying behavior

2.REQUIREMENTS

2.1 FUNCTIONAL REQUIREMENTS



2.1.1 User Account

The user, hereafter referred to as the **customer**, is presented with multiple options as the first step of interaction with the **Recircle Mart Web Application**. The available options depend on whether the user is a **guest user** or a **registered user**, and on the type of activity the user wishes to perform, such as browsing products or making a purchase. A **registered user** is a customer who has already created an account on Recircle Mart by providing personal details and login credentials. This information is stored securely as the user's **profile** in the system database. A registered user can log into the system to access all features, including buying products, selling products, making payments, and viewing order history.

A **new user** can choose Register option to continue:

a) **Register** with the system by providing the required personal information

A registered user may also use the system only for browsing products without logging in, if they do not wish to perform any transactions.

In the context of Recircle Mart, **product availability** refers to viewing product listings, prices, product conditions, and seller details. The system provides the user with the option to exit the application at any stage during interaction.

• 2.1.2 Registration and Creation of User Profile

The system requires a user to **register** in order to perform any transaction-related activities, such as purchasing products, listing items for sale. Registration is not mandatory for browsing or searching products.

During registration, the system collects the following basic information from the user:

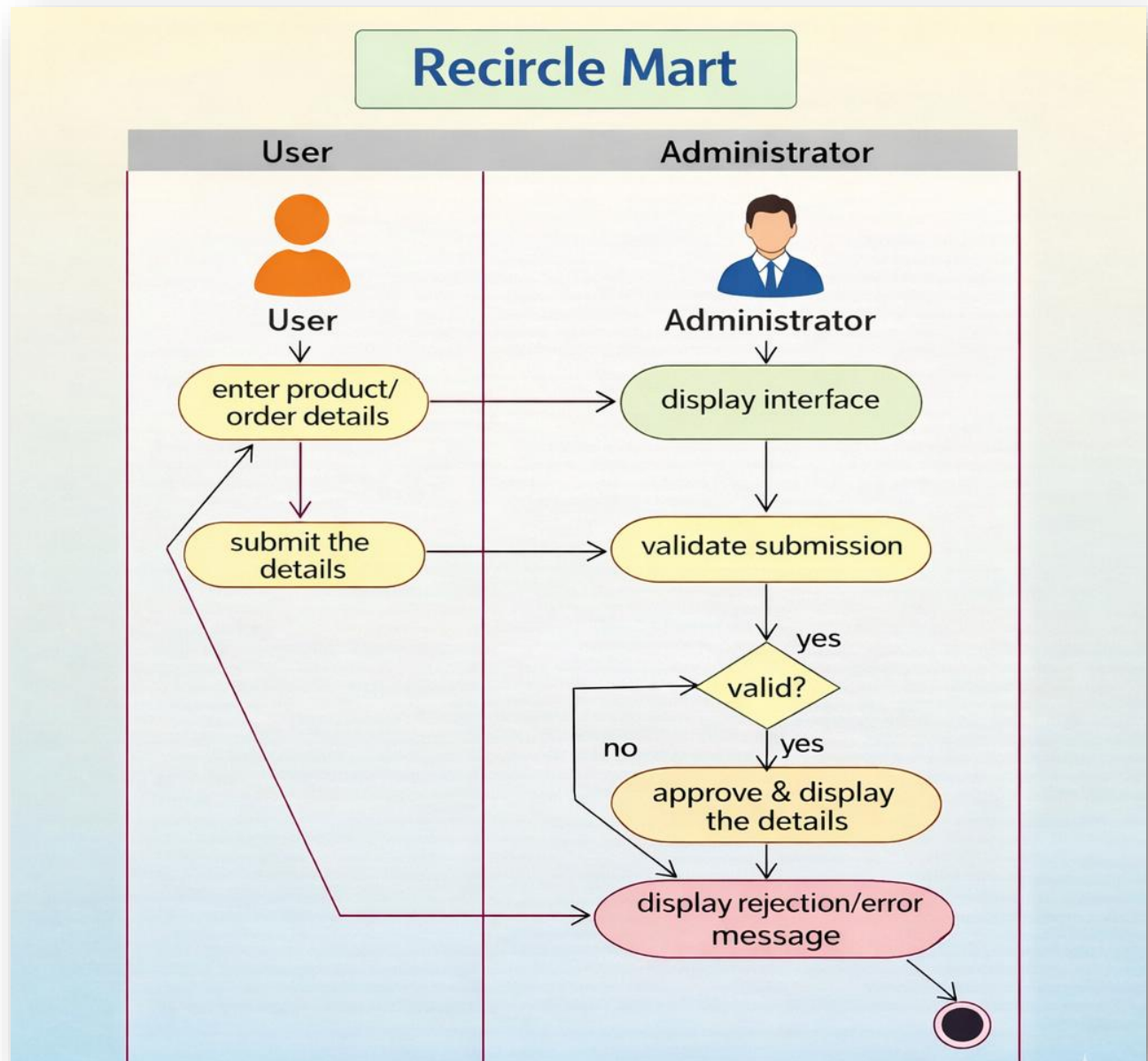
- Username or email ID
- Password
- First name and last name
- Phone number
- Email address

Recircle-Mart Web Application

Upon successful registration, the system automatically creates a **user profile** and stores the details securely in the database. This profile enables the user to:

- Log into the system
- Manage personal information
- View order history
- Track listed products
- Perform secure transactions

The user profile acts as a unique identity for the customer within the **Recircle Mart Web Application** and ensures personalized and secure access to system features.



2.1.3 Quick Search

The Recircle Mart Web Application provides a Quick Search facility that allows any user, including guest users, to browse and search for products without logging into the system. This feature enables users to explore available products and compare prices offered by different sellers on the platform.

Recircle-Mart Web Application

The system allows the user to search products based on various parameters such as:

- Product name
- Product category
- Location
- Price range
- Product condition

The search inputs are entered through user-friendly input fields and dropdown menus. After receiving the search criteria, the system accesses the product database and retrieves all matching product listings.

The search results are displayed in a structured tabular or card-based format showing the following information:

- Product name
- Product category
- Product price
- Product condition
- Seller name
- Product availability

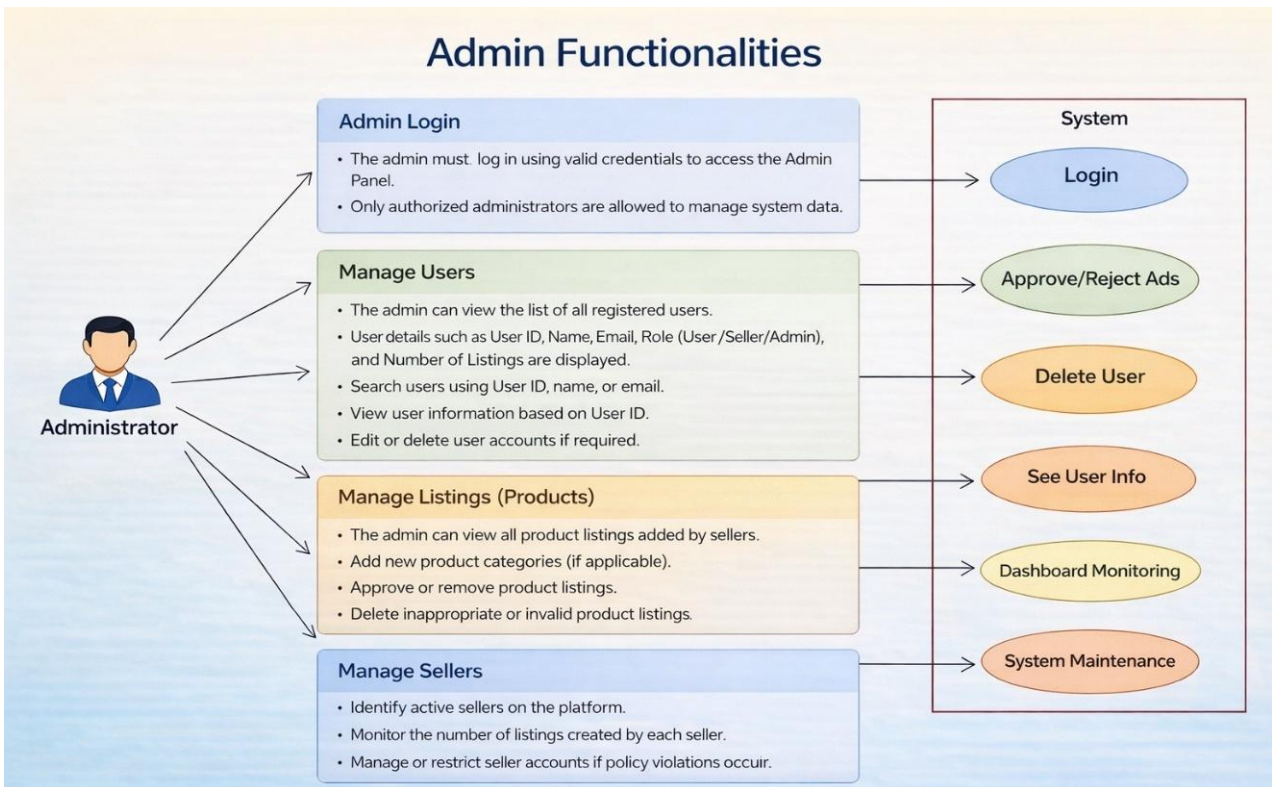
Multiple products from different sellers may be displayed for the same search criteria. Each product listing includes a View Details or Buy Product option.

If no products match the search criteria, the system displays an appropriate message indicating that no results were found.

2.1.4 Buying Products and Making Offer

After viewing product details through Quick Search, the system allows the user to proceed with making offer to the seller.

- a) If the user is a guest user, the system requires the user to register and log in before proceeding further.
- b) If the user is already a registered user and logged in, the system allows the user to directly continue the chat with the seller.



Once the user is authenticated, the system verifies the availability of the selected product. If the product is available, the user is prompted to chat with the seller.

The system then redirects the user to the chat page,

Admin Functionalities

1. Admin Login

- The admin must log in using valid credentials to access the Admin Panel.
- Only authorized administrators are allowed to manage system data.

2. Manage Users

- The admin can view the list of all registered users.
- User details such as **User ID, Name, Email, Role (User/Seller/Admin), and Number of Listings** are displayed.
- The admin can search users using **User ID, name, or email**.
- The admin can view user information based on **User ID**.
- The admin can edit or delete user accounts if required.

Recircle-Mart Web Application

3. Manage Listings (Products)

- The admin can view all product listings added by sellers.
- The admin has the authority to:
 - Add new product categories (if applicable)
 - Approve or remove product listings
 - Delete inappropriate or invalid product listings

4. Manage Sellers

- The admin can identify active sellers on the platform.
- The admin can monitor the number of listings created by each seller.
- Seller accounts can be managed or restricted if policy violations occur.

5. Dashboard Monitoring

- The admin dashboard displays statistical information such as:
 - Total number of users
 - Active sellers
 - Total product listings
 - Recently registered users
- This helps the admin monitor system activity efficiently.

6. System Maintenance

- The admin ensures data integrity and smooth functioning of the application.
- The admin can take corrective actions when errors or misuse are detected

2.2 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality attributes, performance expectations, and operational constraints of the Recircle Mart Web Application.

2.2.1 User Interface

The Recircle Mart Web Application provides a simple, responsive, and user-friendly interface to ensure ease of use for all types of users, including customers, sellers, and administrators.

- The user interface is web-based and accessible through standard web browsers.
- Navigation is intuitive, with clear menus for product search, user management, and transactions.
- The Admin Panel provides dashboards and tables for efficient system management.

Refer to Appendix B for detailed screenshots of user interfaces.

2.2.2 Performance

Number of Concurrent Users

The Recircle Mart system is designed to handle at least 1000 concurrent users or transactions efficiently without significant degradation in performance.

System Reliability

- The application is designed to handle temporary server or network issues gracefully.
 - In case of minor failures, the system ensures data consistency and prevents transaction loss.
 - Proper session handling ensures that user operations are not abruptly interrupted.
-

2.2.3 Constraints

- The system must support a minimum of 1000 concurrent requests or transactions.
 - The application performance depends on server capacity, network availability, and database performance.
 - Security constraints must be followed to protect user data and transaction details.
-

2.2.4 Other Requirements

Hardware Requirements

The system can function efficiently on the following minimum hardware configuration:

- Processor: Intel Dual-Core or equivalent and above
- RAM: 4 GB or higher
- Hard Disk: 20 GB free space or above

Software Requirements

- Operating System: Windows, Linux, or macOS
- Web Server: Apache Tomcat
- Database: MySQL / PostgreSQL / Oracle
- Frontend: HTML, CSS, JavaScript
- Backend: Java / Spring Boot / Node.js (as applicable)
- Browser Support: Google Chrome, Mozilla Firefox, Microsoft Edge

3. DESIGN

3.1 Database Design

Recircle-Mart Database Schema Documentation

This document describes all database tables used in the **Recircle-Mart** database, including their fields, data types, and relationships.

1. users

Column Name	Data Type	Description
id	INT (PK)	Unique user ID
first_name	VARCHAR(100)	User first name
last_name	VARCHAR(100)	User last name
email	VARCHAR(150)	Unique email address
password	VARCHAR(255)	Encrypted password
avatar	TEXT	Profile image path
created_at	TIMESTAMP	Account creation time
role	ENUM	User role (user/admin)
active	Boolean	Active status of user
public_key	VARCHAR	For encrypting the message

2. listings

Column Name	Data Type	Description
id	INT (PK)	Listing ID
title	VARCHAR(255)	Product title
price	INT	Product price
category	VARCHAR(100)	Main category
subcategory	VARCHAR(100)	Sub category
location	TEXT	Product location
year	INT	Purchase year
description	TEXT	Product description
seller_id	INT (FK)	Reference to users(id)
created_at	TIMESTAMP	Created time
status	VARCHAR	Status of product

Recircle-Mart Web Application

3. listing_images

Column Name	Data Type	Description
id	INT (PK)	Image ID
listing_id	INT (FK)	Related listing
image_path	TEXT	Image file path
created_at	TIMESTAMP	Upload time

4. favourites

Column Name	Data Type	Description
id	INT (PK)	Favourite ID
user_id	INT (FK)	User reference
listing_id	INT (FK)	Listing reference
created_at	TIMESTAMP	Added time

5. chats

Column Name	Data Type	Description
id	INT (PK)	Chat ID
listing_id	INT (FK)	Related listing
buyer_id	INT (FK)	Buyer user ID
seller_id	INT (FK)	Seller user ID
created_at	TIMESTAMP	Chat created time

Recircle-Mart Web Application

6. messages

Column Name	Data Type	Description
id	INT (PK)	Message ID
chat_id	INT (FK)	Related chat
sender_id	INT (FK)	Sender user ID
cipherText	TEXT	Encrypted message content
created_at	TIMESTAMP	Sent time
alg	VARCHAR(255)	Algorithm used for encryption
iv	TEXT	Initialization Vector
tag	TEXT	TAG
receiver_id	INT	Receiver ID
enc_key_for_sender	LONGTEXT	Key for decryption for sender

7. notifications

Column Name	Data Type	Description
id	INT (PK)	Notification ID
user_id	INT (FK)	User reference
type	ENUM	Notification type
title	VARCHAR(255)	Notification title
message	TEXT	Notification message
listing_id	INT (FK)	Related listing
related_user_id	INT	Related user
is_read	BOOLEAN	Read status
created_at	TIMESTAMP	Created time

Recircle-Mart Web Application

8. visits

Column Name	Data Type	Description
id	INT (PK)	Visit ID
user_id	INT	User reference (nullable)
ip_address	VARCHAR(100)	Visitor IP
created_at	TIMESTAMP	Visit time

9.Admin_events

Column Name	Data Type	Description
id	INT (PK)	Event ID
title	VARCHAR(255)	Event title
event_date	DATE	Event date
description	TEXT	Event details
created_at	TIMESTAMP	Created time

Recircle-Mart Web Application

Relationships Summary

- users → listings (1 to Many)
 - listings → listing_images (1 to Many)
 - users → favourites (1 to Many)
 - listings → favourites (1 to Many)
 - listings → chats (1 to Many)
 - users → chats (buyer/seller)
 - chats → messages (1 to Many)
 - users → messages (1 to Many)
 - users → notifications (1 to Many)
 - users → visits (1 to Many)
-

4. CODING STANDARDS IMPLEMENTED

The Recircle Mart Web Application follows standard and consistent coding practices to ensure readability, maintainability, scalability, and reusability of the source code. These standards are applied across backend services, frontend components, and database interaction layers.

Naming and Capitalization

The application follows Pascal Case and Camel Case naming conventions as described below.

Identifier	Naming Case	Examples	Additional Notes
Class	PascalCase	UserService, ProductListing, OrderController	Class names represent real-world entities and are nouns. Underscores (<code>_</code>) are not used.
Method	camelCase	createUser(), updateProduct(), getOrderDetails()	Method names use verbs or verb phrases.
Parameter	camelCase	userId, productName	Parameter names are meaningful and self-explanatory.
Interface	PascalCase with “I” prefix	IUserService, IOrderRepository	Interfaces are prefixed with “I” for clear identification.
Property / Variable	camelCase	email, price, orderStatus	Variable names represent the data they store.
Private Member Variable	<code>_camelCase</code>	<code>_password</code> , <code>_totalAmount</code>	Underscore prefix is used for private members.
Exception Class	PascalCase + “Exception”	PaymentException, UserNotFoundException	Exception names clearly indicate error scenarios.

4.2 Commenting Standards

The following commenting standards are followed throughout the Recircle Mart codebase:

- Each class, interface, and non-public method is properly documented.
- End-line comments are used only where necessary, mainly for variable declarations.
- A single space is maintained between the comment delimiter (//) and the comment text.
- All comments begin with an uppercase letter and end with a period.
- Comments explain **why the code exists**, not **what the code does**.
- Redundant or unnecessary comments are avoided to keep the code clean.

5. TEST REPORT

The testing of the Recircle Mart Web Application was conducted by an independent group named Linux.

Both General Testing and Static Testing were performed to ensure correct functionality, usability, and reliability of the system.

GENERAL TESTING:

SR NO	TEST CASE	EXPECTED RESULT	ACTUAL RESULT	ERROR MESSAGE
1	Registration Page	User should be redirected to the next page after successful registration	OK	None
2	Login Page	Pop-up message shown if credentials are incorrect	OK	“Please enter username and password again”
3	Reset Password	Only the user’s password should be reset	OK	None
4	Quick Search Product	Displays all matching product listings	OK	None
5	Product Listing Submission	All mandatory fields must be filled before submission	OK	None
6	Login Status Check	System verifies whether user is logged in or not	OK	None
7	Add Product Details	Allows adding product information correctly	OK	None
8	View Product Page	Displays added product information	OK	None
9	Save Listing Data	Stores all product data in listings table	OK	None
10	Transaction Process	On back navigation, user returns to previous page	OK	None
11	View Transactions	Displays all previous transactions	OK	None
12	Logout	User is logged out successfully	OK	None

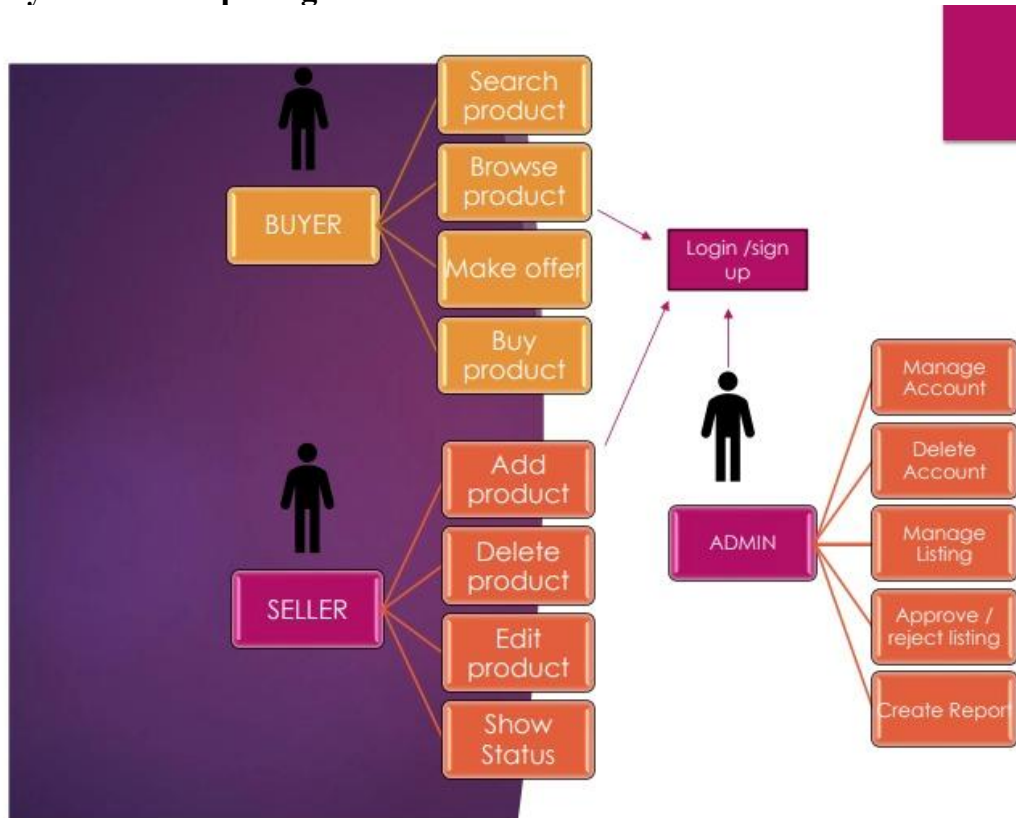
6. PROJECT MANAGEMENT RELATED STATISTICS

Date	Work Performed	SDLC Phase	Additional Notes
Oct 01, 2025	Project allotment and requirement gathering	Feasibility Study	Initial discussions were conducted to understand the scope and objectives of the Recircle Mart Web Application.
oct 20, 2025	Preparation and validation of SRS document and team role allocation	Requirement Analysis	Functional and non-functional requirements were finalized.
oct 30, 2025	Design of Use Case Diagram, ER Diagram, Class Diagram, and UI mockups	System Design	Database schema and system architecture were completed.
nov 15, 2025	Business logic and module design started	Design Phase	Core modules such as User, Product, Order, and Admin were identified.
dec 05, 2025	Coding phase started	Implementation	Backend and frontend development initiated.
dec 26, 2025	Development of user and seller modules	Implementation	User authentication and product listing modules implemented.
jan 07, 2026	Continued development and unit testing	Implementation & Unit Testing	Bugs identified and resolved during testing.
jan 20, 2026	Integration of payment and admin modules	Integration Testing	Payment flow and admin dashboard tested successfully.
jan 24, 2026	System testing by project team	Testing (Module Testing)	All modules tested individually.
jan 26, 2026	Acceptance testing by peer group	Testing (Acceptance Testing)	Feedback received and minor improvements applied.
jan 30, 2026	Debugging and final refinements	Debugging	Application stabilized and optimized.
Feb 03, 2026	Final project submission	Deployment	Recircle Mart Web Application submitted successfully.

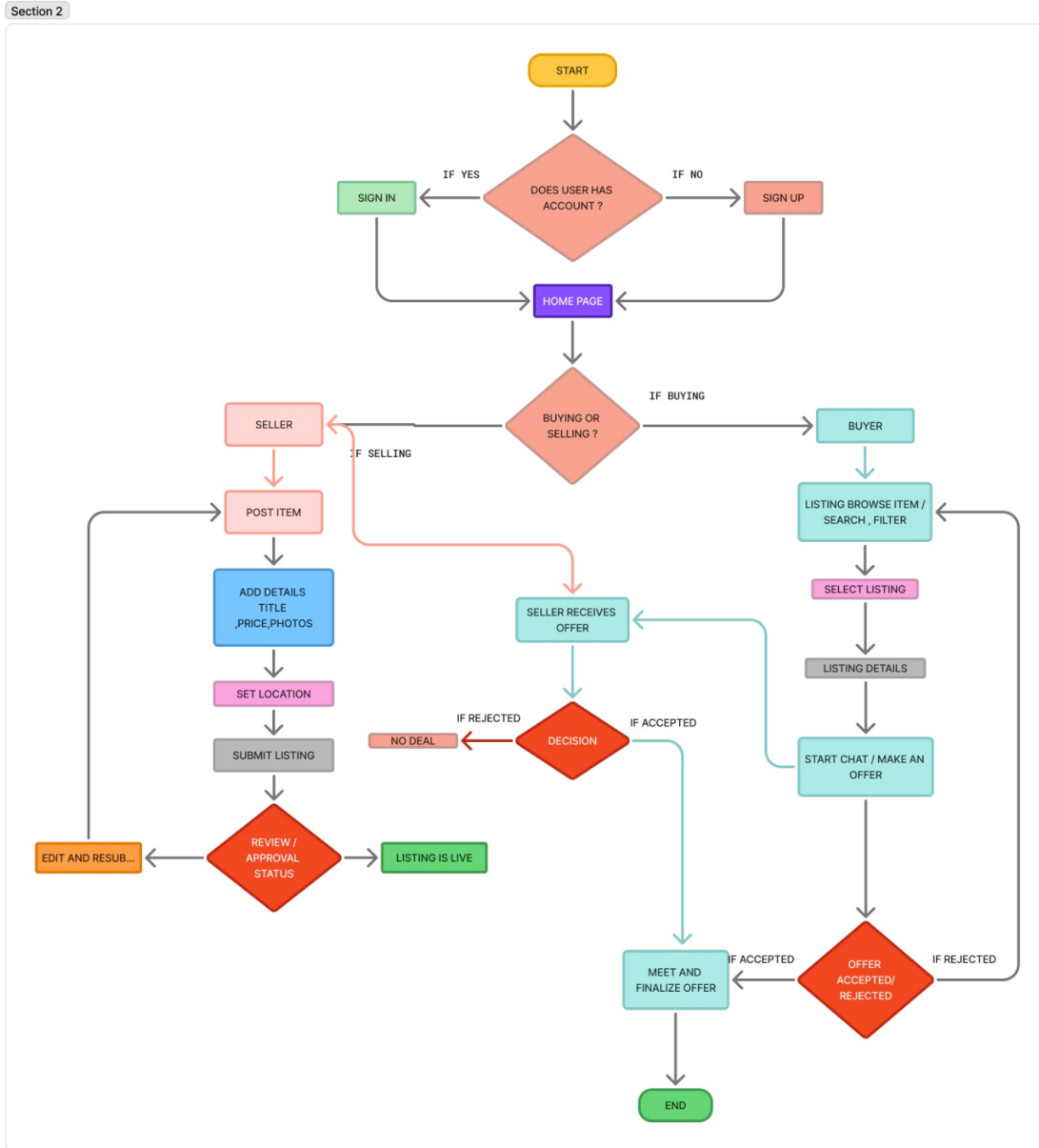
E-R Diagram,Dataflow diagram and Class Diagram:

Appendix A

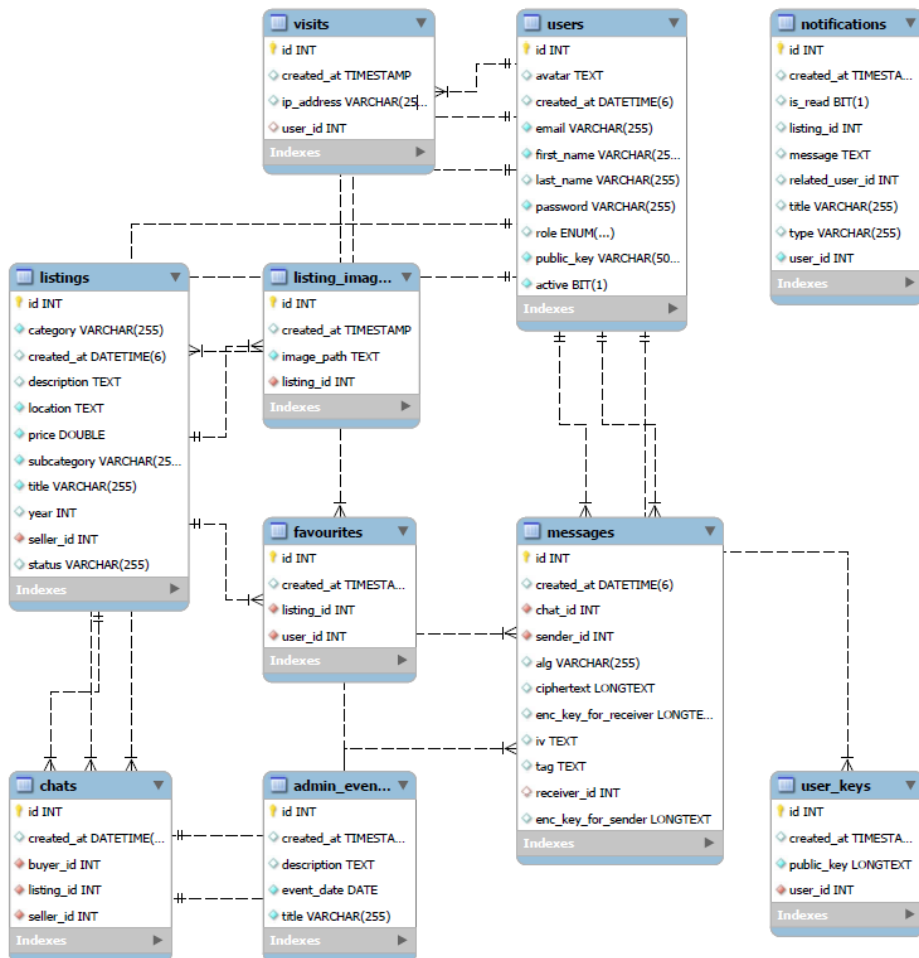
Entity Relationship Diagram



Data Flow Diagram:

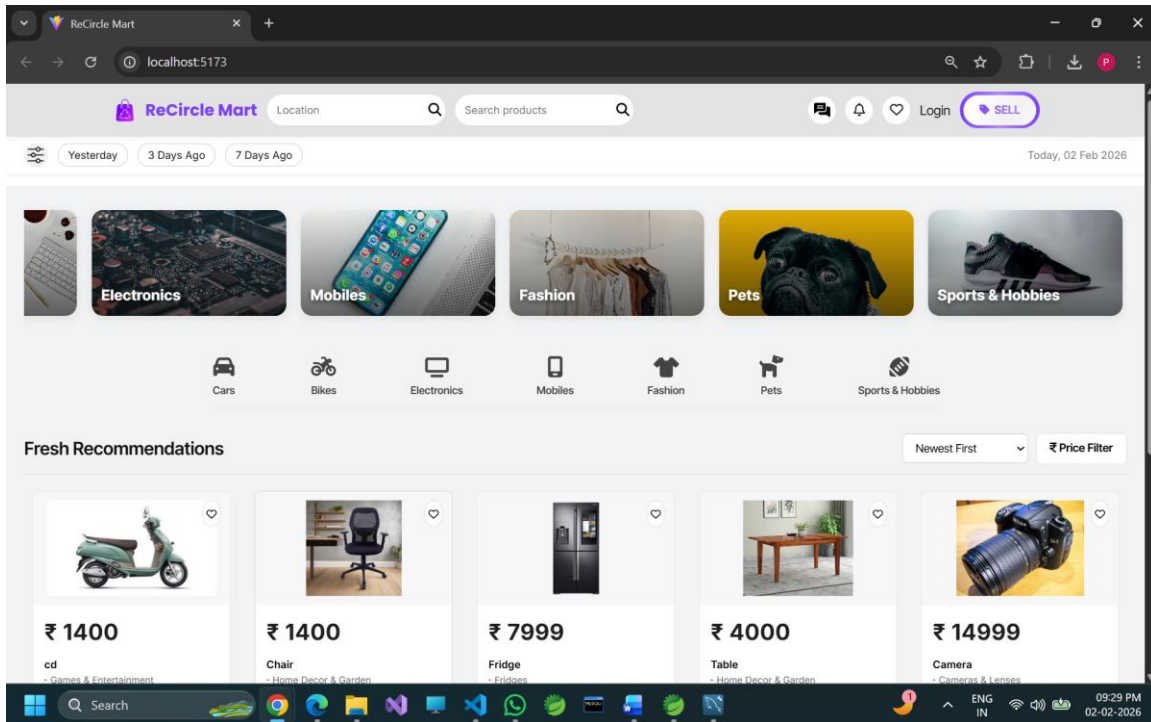


Class Diagram

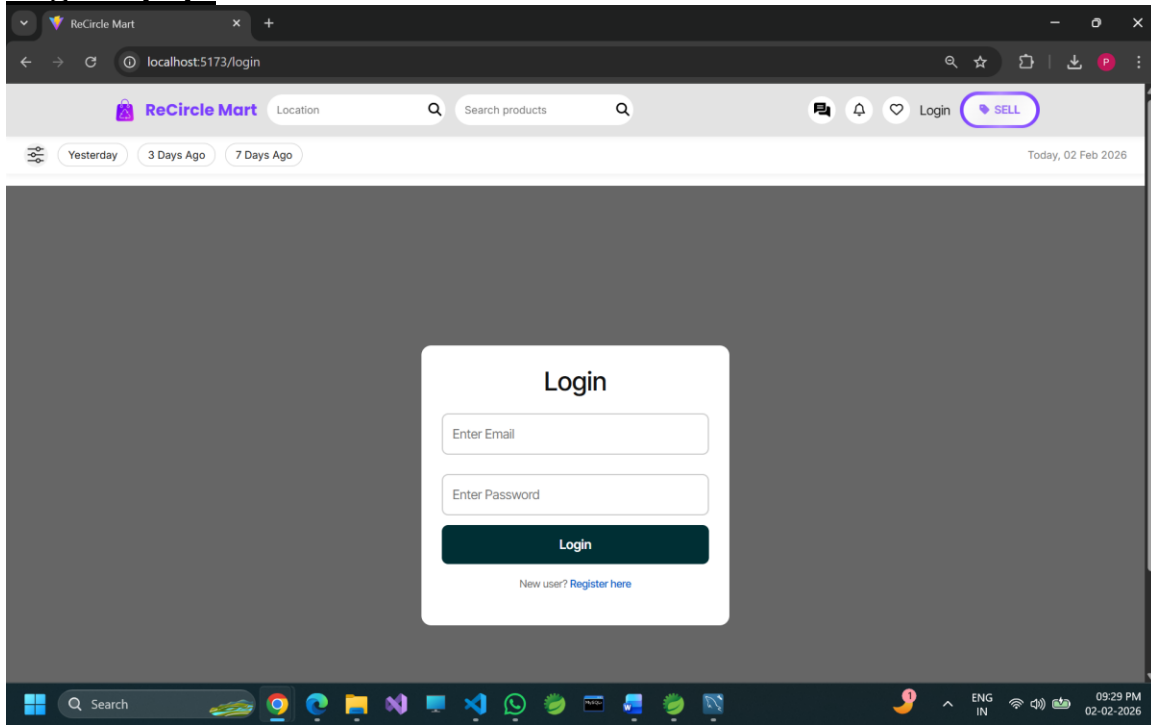


Appendix B

Homepage:



LoginPopup:



Recircle-Mart Web Application

AdministratorPage:

The screenshot displays the Recircle-Mart Admin Panel. The left sidebar contains navigation links: Dashboard, Manage Listings, Manage Users, and Calendar. The main content area is titled 'Manage Users' and features four summary cards: Total Users (5), Active Sellers (3), Total Listings (8), and Recent Users (30 days) (0). Below these cards is a search bar and tabs for 'All Users', 'Normal Users', 'Sellers', and 'Admins'. A table lists users with columns for Profile, ID, Name, Email, Role, Listings, and Actions. The table shows five users, including administrators and sellers. At the bottom, there is a 'Logout' button and pagination controls showing 'Page 1 of 1'.

Profile	ID	Name	Email	Role	Listings	Actions
	7		abc@gmail.com	Seller	1	
	4		test2@gmail.com	User	0	
	3		harsh@gmail.com	Seller	2	
	2		rohit@gmail.com	Admin	0	
	1		prashant@gmail.com	Admin	5	

User_Registration:

The screenshot shows the Recircle-Mart User Registration page. The header includes the site name, a search bar, and links for Login and Sell. The main content area features a 'Register' form with a 'Choose Profile Icon' section displaying a grid of 16 icons. Below the icons is a link to 'Or upload your own photo'. The form includes input fields for First Name, Last Name, Email, and Password, followed by a 'Register' button. At the bottom, there is a link for users who are already registered to 'Login here'.

Recircle-Mart Web Application

Sell Product

The screenshot shows the 'Post your Ad' form in the Recircle-Mart web application. The form is titled 'Post your Ad' and includes the following sections:

- Selected category:** Furniture (with a 'Change' link).
- Include some details:**
 - Sub Category *:** A dropdown menu with 'Select' as the current selection.
 - Title *:** A text input field.
 - Year of Purchase *:** A text input field.
 - Description *:** A text area.
- Set a price:** A text input field with a rupee symbol (₹) as a prefix.
- Upload up to 12 photos:** A section with four placeholder images, each with a '+' icon for uploading.

The browser address bar shows 'localhost:5173/sell/furniture'. The user is logged in as 'Prashant' and has a 'SELL' button in the top right corner. The Windows taskbar at the bottom shows the time as 09:35 PM on 02-02-2026.

User Dashboard

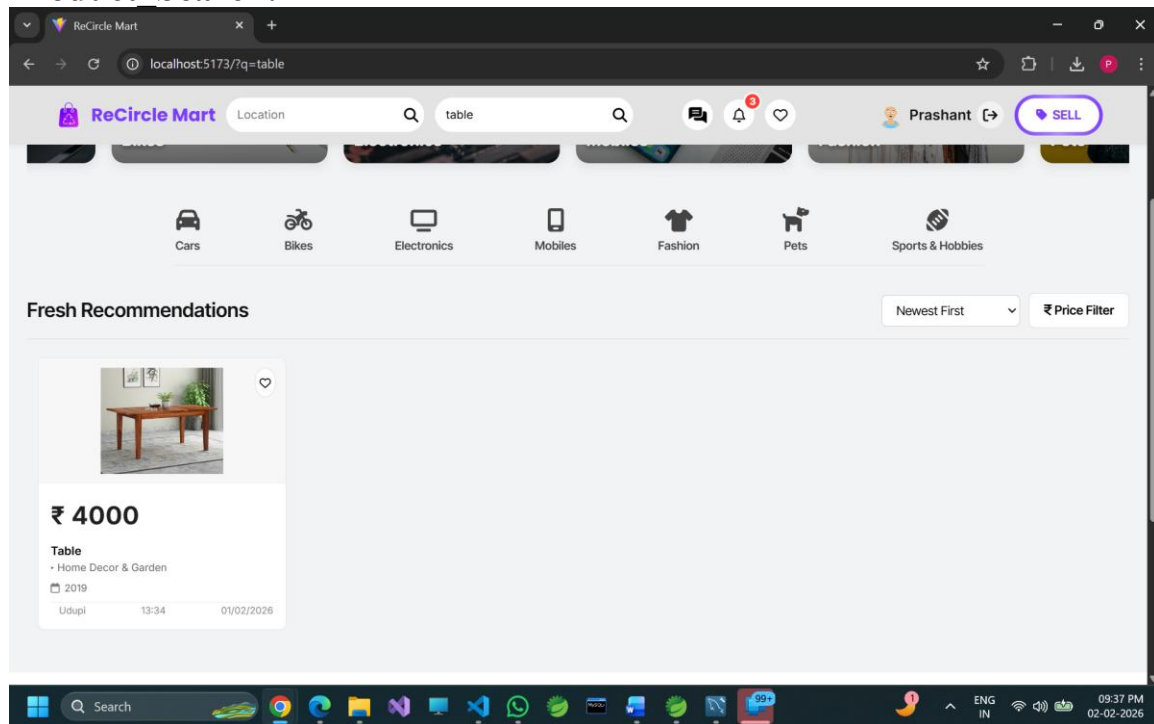
The screenshot shows the 'User Dashboard' in the Recircle-Mart web application. The dashboard includes the following elements:

- Header:** Recircle-Mart logo, location search, product search, user profile (Prashant), and a 'SELL' button.
- Filters:** 'Yesterday', '3 Days Ago', and '7 Days Ago' buttons.
- Today, 02 Feb 2026:** A date indicator.
- My Listings:** A section displaying the user's listings with a profile card and a grid of items.
 - Profile Card:** Prashant Kumar, prashant@gmail.com, 5 Listings, and an 'Edit Profile' button.
 - Listings Grid:**
 - Table:** ₹ 4000, with a 'View' button.
 - Camera:** ₹ 14999, with a 'View' button.
 - Bike:** ₹ 8000, with a 'View' button.
 - TV:** ₹ 3000, with a 'View' button.
 - Item:** ₹ 13998, with a 'View' button.

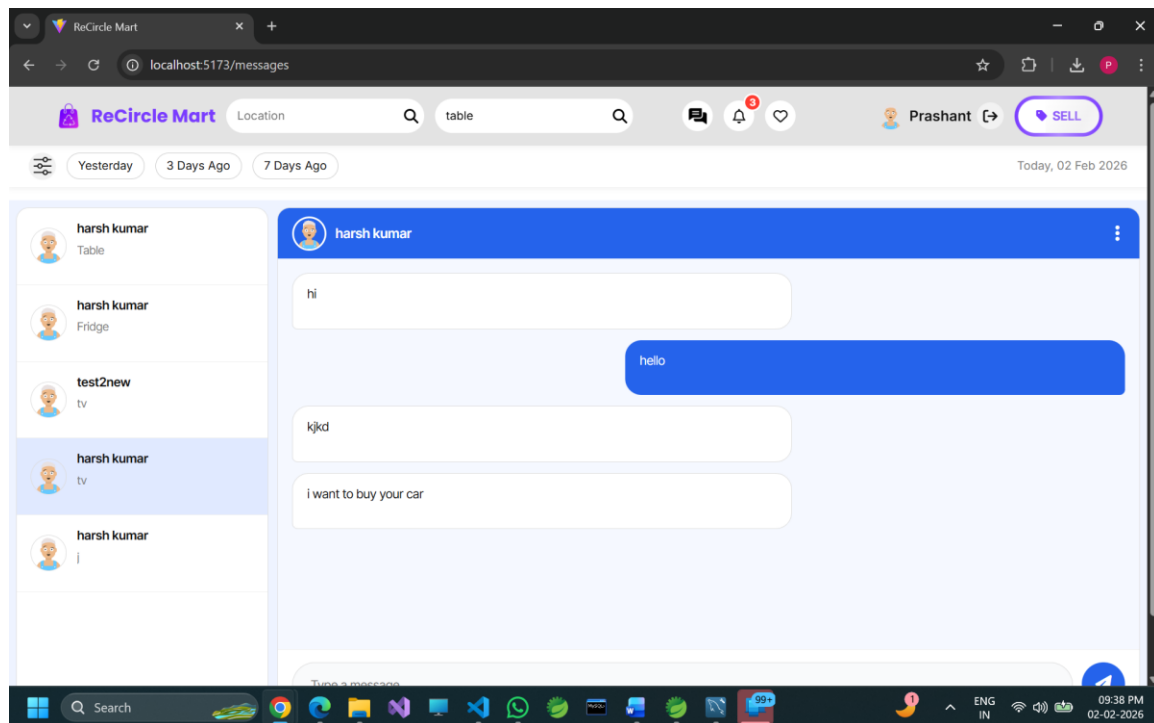
The browser address bar shows 'localhost:5173/profile'. The Windows taskbar at the bottom shows the time as 09:36 PM on 02-02-2026.

Recircle-Mart Web Application

Product Search:



Chat service:



Recircle-Mart Web Application

Edit Profile:

ReCircle Mart

localhost:5173/edit-profile

ReCircle Mart

Location

table

Prashant

SELL

Edit Profile

Update your personal information

Name

Prashant

Email

prashant@gmail.com

New Password

Leave empty to keep current password

Save Changes

Cancel

09:39 PM 02-02-2026

Product Details:

ReCircle Mart

localhost:5173/product/50

ReCircle Mart

Location

table

Prashant

SELL

Chair

Furniture / Home Decor & Garden

₹ 1400

Make Offer

Overview

2023

02/02/2026

harsh

Member since recently

Chat with seller

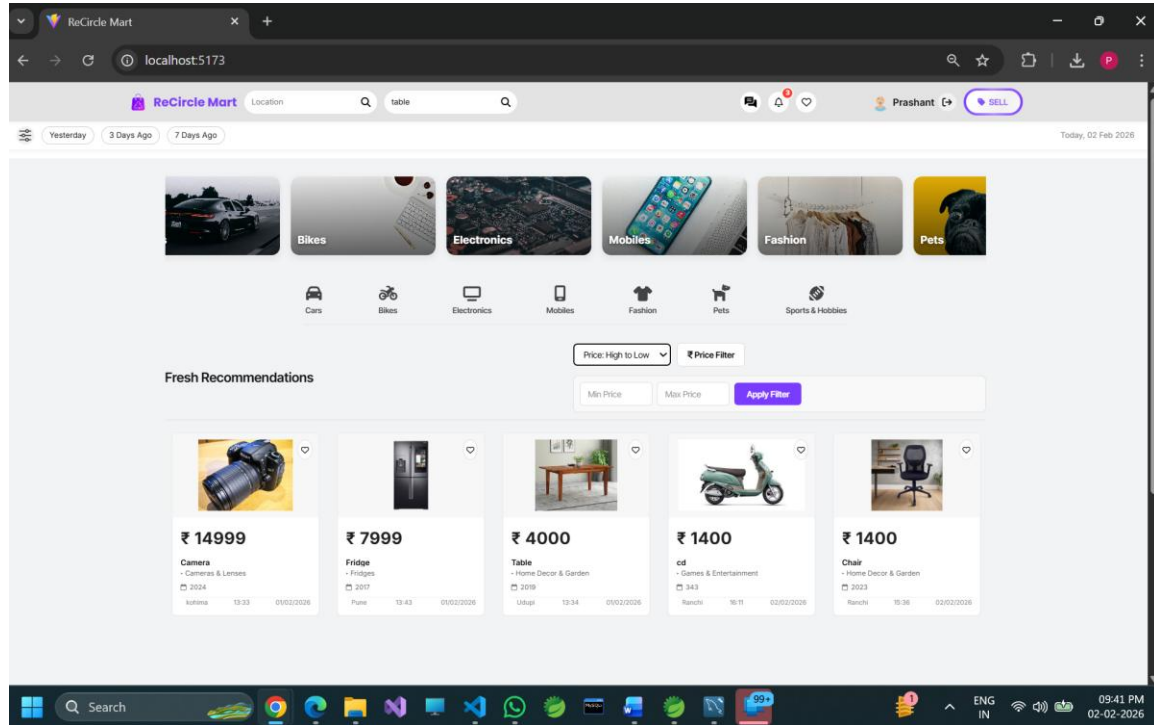
Description

Notes: chair

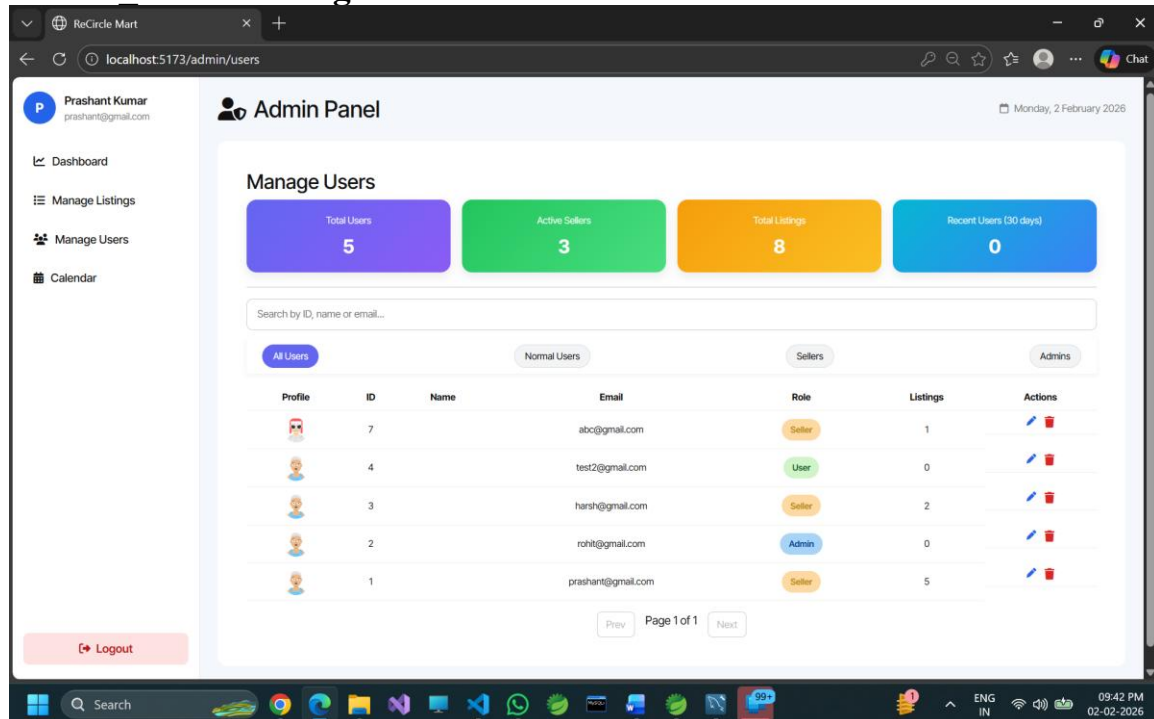
09:40 PM 02-02-2026

Recircle-Mart Web Application

ProductFilters:



Admin Panel-ManageUsers:



7.REFERENCES:

1. <https://www.google.com>
(General reference and technical research)
2. <https://www.olx.in>
(Reference for buy-and-sell marketplace functionality)
3. <https://www.quikr.com>
(Online classified advertisement platform reference)
4. <https://developer.mozilla.org>
(Frontend and backend web development documentation)
5. <https://www.w3.org>
(Web standards for HTML, CSS, and JavaScript)
6. <https://spring.io>
(Spring Boot framework reference for backend development)
7. <https://www.mysql.com>
(Database management system reference)
8. <https://www.wikipedia.org>
(Conceptual reference for system design and architecture)

A PROJECT ON
“RECIRCLE- MART WEB APPLICATION”

SUBMITTED IN
PARTIAL FULFILLMENT OF THE REQUIREMENT
FOR THE COURSE OF
DIPLOMA IN ADVANCED COMPUTING FROM CDAC



SUNBEAM INSTITUTE OF INFORMATION TECHNOLOGY
Hinjawadi

SUBMITTED BY:

Prashant Kumar
Rohit Kavathekar
Harsh Kumar
Om Dhavale

UNDER THE GUIDENCE OF:

Vrushabh Kumatale
Faculty Member
Sunbeam Institute of Information Technology, PUNE.

CERTIFICATE

This is to certify that the project work under the title “ReCircle-Mart” Web Application is done by Prashant Kumar, Rohit Kavathekar, Harsh Kumar, Om Dhavale in partial fulfillment of the requirement for award of Diploma in Advanced Computing Course.

Project Guide
Vrushabh Kumatale

Mr. Yogesh Kolhe
Course Co-Coordinator

Date:03/02/2026

